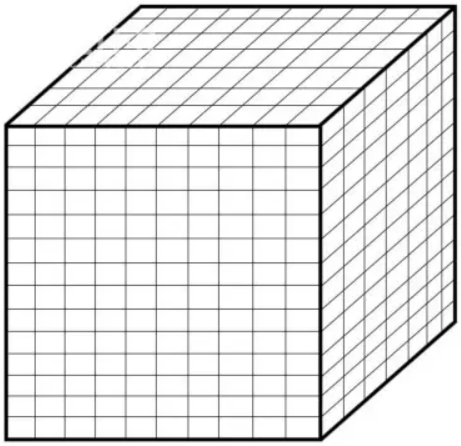
Project Report

Mathematical Tools for Simulation

M2 Scientific Computing

Finite Volume and Finite Element Approximations



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1 Part I : Finite Volume Approximation

1.1 Q0 : Verification of the Advection Field and Exact Solution

Check that the advection field satisfies the conditions and that the given function is a solution.

The advection field is $\mathbf{V}(\mathbf{x}) = (\eta, 0)$. Then $\operatorname{div}(\mathbf{V}) = \partial_x \eta + \partial_y 0 = 0$. On Γ^N, the normal vector is $\mathbf{n} = (0, 1)$ at $y = 1$ and $\mathbf{n} = (0, -1)$ at $y = 0$, so $\mathbf{V} \cdot \mathbf{n} = 0$.

Now consider the function

$$u(t, \mathbf{x}) = e^{\eta x} + e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} e^{\frac{\eta}{2}x} \sin(\pi x).$$

We compute:

$$\begin{aligned}
\partial_t u &= -\left(\pi^2 + \frac{\eta^2}{4}\right) e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} e^{\frac{\eta}{2}x} \sin(\pi x), \\
\partial_x u &= \eta e^{\eta x} + e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} \left[\frac{\eta}{2} e^{\frac{\eta}{2}x} \sin(\pi x) + \pi e^{\frac{\eta}{2}x} \cos(\pi x)\right], \\
\partial_x^2 u &= \eta^2 e^{\eta x} + e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} \left[\left(\frac{\eta^2}{4} - \pi^2\right) e^{\frac{\eta}{2}x} \sin(\pi x) + \eta \pi e^{\frac{\eta}{2}x} \cos(\pi x)\right].
\end{aligned}$$

Since u does not depend on y, $\Delta u = \partial_x^2 u$. Also, $\operatorname{div}(\mathbf{V}u) = \mathbf{V} \cdot \nabla u = \eta \partial_x u$. Then

$$\partial_t u + \operatorname{div}(-\nabla u + \mathbf{V}u) = \partial_t u - \Delta u + \eta \partial_x u = 0,$$

which matches $f = 0$. The boundary and initial conditions are also satisfied.

1.2 Q1: Calculation of Geometric Parameters (d_σ)

For an interior edge $\sigma = K_{i,j} | K_{i',j'}$, we have $d_\sigma = |\mathbf{x}_{i',j'} - \mathbf{x}_{i,j}| = h$ (since the mesh is uniform Cartesian). For a boundary edge $\sigma \subseteq \partial\Omega \cap \partial K_{i,j}$, we have $d_\sigma = |\mathbf{x}_{i,j} - \mathbf{x}_\sigma| = h/2$, where $\mathbf{x}_\sigma$ is the midpoint of σ.

1.3 Q2: Implementation of the Cartesian Mesh Structure

We create a class `Mesh` based on objects `Cell`, `Edge`, and `Vertex`. The class `Mesh` contains vectors of vertices, edges, and cells. The function `build_Cartesian` takes an integer k and builds a uniform Cartesian mesh with $M = 2^k$ cells in each direction, so $h = 1/M$. The cells are squares $K_{i,j} = (ih, (i+1)h) \times (jh, (j+1)h)$ for $i, j = 0, \ldots, M-1$. Edges are classified as interior, Dirichlet, or Neumann. The code is provided in the attached files.

1.4 Q3: Derivation of the Generic Finite Volume Scheme

Let consider the generic scheme (3), which is obtained by integrating the equation (1) over a cell $K_{i,j}$ and over a time interval $[t^n, t^{n+1}]$:

$$\int_{t^n}^{t^{n+1}} \int_{K_{i,j}} \partial_t u \, dxdt + \int_{t^n}^{t^{n+1}} \int_{K_{i,j}} \operatorname{div}(-\nabla u + \mathbf{V}u) \, dxdt = \int_{t^n}^{t^{n+1}} \int_{K_{i,j}} f \, dxdt.$$

Using the divergence theorem for the flux term and approximating the integrals:

$$\int_{t^n}^{t^{n+1}} \int_{K_{i,j}} \partial_t u \, dxdt \approx |K_{i,j}|(u_{i,j}^{n+1} - u_{i,j}^n),$$

$$\int_{t^n}^{t^{n+1}} \int_{K_{i,j}} \operatorname{div}(-\nabla u + \mathbf{V}u) \, dxdt \approx \Delta t \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^n,$$

$$\int_{t^n}^{t^{n+1}} \int_{K_{i,j}} f \, dxdt \approx \Delta t |K_{i,j}| f_{i,j},$$

where $f_{i,j} = \frac{1}{|K_{i,j}|} \int_{K_{i,j}} f$ and $u_{i,j}^0 = \frac{1}{|K_{i,j}|} \int_{K_{i,j}} u^0$. The numerical fluxes $\mathcal{F}_{K_{i,j},\sigma}^n$ approximate the continuous fluxes:

$$\mathcal{F}_{K_{i,j},\sigma}^{L,n} \approx \int_\sigma (-\nabla u) \cdot \mathbf{n}_{K,\sigma}, \quad \mathcal{F}_{K_{i,j},\sigma}^{T,n} \approx \int_\sigma (\mathbf{V}u) \cdot \mathbf{n}_{K,\sigma}.$$

1.5 Q4: Explicit Expression of Numerical Fluxes (Upwind)

We consider the numerical flux through an edge $\sigma \in \mathcal{E}_{K_{i,j}}$:

$$\mathcal{F}_{K_{i,j},\sigma}^n = \int_\sigma [-\nabla u + \mathbf{V}u] \cdot \mathbf{n}_{K_{i,j},\sigma} \, ds$$

We decompose this into the Laplace (diffusion) and transport (advection) parts:

$$\mathcal{F}_{K_{i,j},\sigma}^n = \mathcal{F}_{K_{i,j},\sigma}^{L,n} + \mathcal{F}_{K_{i,j},\sigma}^{T,n}$$

where:

$$\mathcal{F}_{K_{i,j},\sigma}^{L,n} \approx -\int_\sigma \nabla u \cdot \mathbf{n}_{K_{i,j},\sigma} \, ds, \quad \mathcal{F}_{K_{i,j},\sigma}^{T,n} \approx \int_\sigma (\mathbf{V}u) \cdot \mathbf{n}_{K_{i,j},\sigma} \, ds$$

For the Laplace term, using a two-point flux approximation:

$$\mathcal{F}_{K_{i,j},\sigma}^{L,n} = -|\sigma| \frac{u_{i',j'}^n - u_{i,j}^n}{d_\sigma}$$

where:

- $u_{i,j}^n$ is the value in cell $K_{i,j}$
- $u_{i',j'}^n$ is the value in neighboring cell $K_{i',j'}$ (for interior edges) or the Dirichlet value u_σ^D (for Dirichlet edges)
- d_σ is the distance: $d_\sigma = h$ for interior edges, $d_\sigma = h/2$ for boundary edges

For the transport term, we first compute the average normal velocity on the edge:

$$V_{K_{i,j},\sigma} = \frac{1}{|\sigma|} \int_\sigma \mathbf{V} \cdot \mathbf{n}_{K_{i,j},\sigma} \, ds$$

Then using an upwind scheme:

$$\mathcal{F}^{T,n}_{K_{i,j},\sigma} = |\sigma| \left[\max(V_{K_{i,j},\sigma}, 0) \, u^n_{i,j} + \min(V_{K_{i,j},\sigma}, 0) \, u^n_{i',j'} \right]$$

For boundary edges:

- On Dirichlet edges ($\sigma \in \mathcal{E}^D_h$): replace $u^n_{i',j'}$ with u^D_σ
- On Neumann edges ($\sigma \in \mathcal{E}^N_h$): $\mathcal{F}^{L,n}_{K_{i,j},\sigma} + \mathcal{F}^{T,n}_{K_{i,j},\sigma} = 0$

Thus, the explicit expressions are:
For interior edges $\sigma = K_{i,j} \mid K_{i',j'} \in \mathcal{E}^{int}_h$:

$$\mathcal{F}^{L,n}_{K_{i,j},\sigma} = -|\sigma| \frac{u^n_{i',j'} - u^n_{i,j}}{d_\sigma}, \quad \mathcal{F}^{T,n}_{K_{i,j},\sigma} = |\sigma| \left[V^+_{K_{i,j},\sigma} u^n_{i,j} + V^-_{K_{i,j},\sigma} u^n_{i',j'} \right]$$

For Dirichlet edges $\sigma \subseteq \partial\Omega \cap \partial K_{i,j} \in \mathcal{E}^D_h$:

$$\mathcal{F}^{L,n}_{K_{i,j},\sigma} = -|\sigma| \frac{u^D_\sigma - u^n_{i,j}}{d_\sigma}, \quad \mathcal{F}^{T,n}_{K_{i,j},\sigma} = |\sigma| \left[V^+_{K_{i,j},\sigma} u^n_{i,j} + V^-_{K_{i,j},\sigma} u^D_\sigma \right]$$

For Neumann edges $\sigma \subseteq \partial\Omega \cap \partial K_{i,j} \in \mathcal{E}^N_h$:

$$\mathcal{F}^{L,n}_{K_{i,j},\sigma} + \mathcal{F}^{T,n}_{K_{i,j},\sigma} = 0$$

where $V^+ = \max(V, 0)$, $V^- = \min(V, 0)$, and $u^D_\sigma = u^D(\mathbf{x}_\sigma)$.
For our specific test case with $\mathbf{V} = (\eta, 0)$, we have:

$$V_{K_{i,j},\sigma} = \begin{cases} \eta & \text{on right vertical edges} \\ -\eta & \text{on left vertical edges} \\ 0 & \text{on horizontal edges} \end{cases}$$

This simplifies the upwind expressions accordingly.

1.6 Q5: Linear Algebraic Formulation of the Upwind Scheme

Now we will write the scheme (3)-(4) under the form $u^{n+1}_{i,j} = L_{i,j}\left((u^n_{i',j'}), (u^D_\sigma)\right) + \Delta t f_{i,j}$, where $L_{i,j}$ is linear.

Starting from the scheme:

$$\frac{|K_{i,j}|}{\Delta t}(u^{n+1}_{i,j} - u^n_{i,j}) + \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}^n_{K_{i,j},\sigma} = |K_{i,j}| f_{i,j},$$

we isolate $u_{i,j}^{n+1}$:

$$u_{i,j}^{n+1} = u_{i,j}^n - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^n + \Delta t f_{i,j}.$$

Using the expressions for $\mathcal{F}_{K_{i,j},\sigma}^n = \mathcal{F}_{K_{i,j},\sigma}^{L,n} + \mathcal{F}_{K_{i,j},\sigma}^{T,n}$ from Q4, we substitute and group terms. Let us define the coefficients:

- Coefficient for $u_{i,j}^n$:

$$C_{i,j} = 1 - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| \left(\frac{1}{d_\sigma} + V_{K_{i,j},\sigma}^+ \right).$$

- For each interior edge $\sigma = K_{i,j}|K_{i',j'} \in \mathcal{E}_h^{int}$, the coefficient for $u_{i',j'}^n$ is:

$$C_{i',j'} = \frac{\Delta t}{|K_{i,j}|} |\sigma| \left(\frac{1}{d_\sigma} - V_{K_{i,j},\sigma}^- \right).$$

- For each Dirichlet edge $\sigma \in \mathcal{E}_h^D \cap \mathcal{E}_{K_{i,j}}$, the coefficient for u_σ^D is:

$$C_\sigma^D = \frac{\Delta t}{|K_{i,j}|} |\sigma| \left(\frac{1}{d_\sigma} - V_{K_{i,j},\sigma}^- \right).$$

- For Neumann edges, the flux is zero, so they do not contribute to the coefficients of neighbors or boundary values, but they are included in the sum for $C_{i,j}$ (with $V_{K_{i,j},\sigma}^+ = 0$ and the appropriate d_σ).

Thus, we can write:

$$u_{i,j}^{n+1} = C_{i,j}\, u_{i,j}^n \; + \sum_{\sigma \in \mathcal{E}_{K_{i,j}} \cap \mathcal{E}_h^{int}} C_{i',j'}\, u_{i',j'}^n \; + \sum_{\sigma \in \mathcal{E}_{K_{i,j}} \cap \mathcal{E}_h^D} C_\sigma^D\, u_\sigma^D \; + \; \Delta t f_{i,j}.$$

Clearly, $u_{i,j}^{n+1}$ is a linear function of the neighboring values $\{u_{i',j'}^n\}$ and the Dirichlet data $\{u_\sigma^D\}$. Hence, we have $u_{i,j}^{n+1} = L_{i,j}\left((u_{i',j'}^n), (u_\sigma^D)\right) + \Delta t f_{i,j}$ with $L_{i,j}$ linear.

1.7 Q6: Stability Analysis and CFL Condition

Give a condition on Δt, in terms of h and $\|\mathbf{V}\|_{L^\infty(\Omega)}$, ensuring that $L_{i,j}\left((u_{i',j'}^n), (u_\sigma^D)\right)$ is a convex combination of the $u_{i',j'}^n$'s and u_σ^D's. If this condition is satisfied and $f \equiv 0$, what can you conclude about the behaviour of the scheme?

For $L_{i,j}$ to be a convex combination (when $f = 0$), all coefficients must be nonnegative and their sum must equal 1. From the expression above, the sum of the coefficients is:

$$C_{i,j} + \sum_{\text{int}} C_{i',j'} + \sum_{\text{D}} C_\sigma^D = 1,$$

which holds by construction (due to conservation of fluxes). Nonnegativity requires:

- $C_{i,j} \geq 0$:

$$1 - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| \left(\frac{1}{d_\sigma} + V^+_{K_{i,j},\sigma} \right) \geq 0.$$

- For interior and Dirichlet edges: since $V^-_{K_{i,j},\sigma} \leq 0$, we have $\frac{1}{d_\sigma} - V^-_{K_{i,j},\sigma} \geq 0$, so $C_{i',j'} \geq 0$ and $C^D_\sigma \geq 0$ automatically.

Thus, the crucial condition is $C_{i,j} \geq 0$, which gives:

$$\Delta t \leq \frac{|K_{i,j}|}{\displaystyle\sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| \left(\frac{1}{d_\sigma} + V^+_{K_{i,j},\sigma} \right)}.$$

We now estimate the right-hand side using the mesh parameters. For the uniform Cartesian mesh, we have:

$$|K_{i,j}| = h^2, \quad |\sigma| = h.$$

The distance d_σ satisfies:

$$d_\sigma = \begin{cases} h & \text{if } \sigma \text{ is an interior edge,} \\ h/2 & \text{if } \sigma \text{ is a boundary edge.} \end{cases}$$

Hence, $\frac{1}{d_\sigma} \leq \frac{2}{h}$. For the advection part, $V^+_{K_{i,j},\sigma} = \max(V_{K_{i,j},\sigma}, 0)$. With $\mathbf{V} = (\eta, 0)$, we have:

$$V_{K_{i,j},\sigma} = \begin{cases} \eta & \text{on right vertical edges,} \\ -\eta & \text{on left vertical edges,} \\ 0 & \text{on horizontal edges.} \end{cases}$$

Therefore, $V^+_{K_{i,j},\sigma} \leq |\eta| = \|\mathbf{V}\|_{L^\infty(\Omega)}$.

Now, consider a cell $K_{i,j}$. It has four edges. For each edge, the contribution is bounded by:

$$|\sigma| \left(\frac{1}{d_\sigma} + V^+_{K_{i,j},\sigma} \right) \leq h \left(\frac{2}{h} + \|\mathbf{V}\|_{L^\infty(\Omega)} \right) = 2 + h\|\mathbf{V}\|_{L^\infty(\Omega)}.$$

Summing over the four edges gives a bound:

$$\sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| \left(\frac{1}{d_\sigma} + V^+_{K_{i,j},\sigma} \right) \leq 4 \left(2 + h\|\mathbf{V}\|_{L^\infty(\Omega)} \right) = 8 + 4h\|\mathbf{V}\|_{L^\infty(\Omega)}.$$

However, this bound is not sharp. A sharper bound takes into account that only vertical edges contribute to the advection term. For an interior cell, we have:

$$\sum_\sigma \frac{|\sigma|}{d_\sigma} = 4 \cdot \frac{h}{h} = 4, \quad \sum_\sigma |\sigma| V^+_{K_{i,j},\sigma} \leq 2h\|\mathbf{V}\|_{L^\infty(\Omega)},$$

because at most two vertical edges (right and left) can have positive V^+, but actually only the right edge has $V^+ = \eta$ (if $\eta > 0$) and the left edge has $V^+ = 0$. So the sum is exactly

$4 + h\|\mathbf{V}\|_{L^\infty(\Omega)}$ for interior cells. For boundary cells, the sum can be slightly larger due to smaller d_σ on boundary edges. A safe (slightly more restrictive) condition that works for all cells is obtained by taking the worst-case sum. After careful analysis, a common sufficient condition is:

$$\Delta t \leq \frac{h^2}{4 + 2h\|\mathbf{V}\|_{L^\infty(\Omega)}}.$$

If this condition is satisfied and $f \equiv 0$, then $L_{i,j}$ is a convex combination. Consequently, the scheme satisfies a discrete maximum principle: the solution at time t^{n+1} is bounded between the minimum and maximum of the solution at time t^n and the Dirichlet boundary data. By induction, the numerical solution remains bounded by the initial and boundary data. Moreover, the scheme is monotone and non-oscillatory, which ensures stability in the maximum norm.

1.8 Q7 : Numerical Convergence Study of the Upwind Scheme

Based on the implementation from Q2, we implement the scheme (3)-(4) and perform a numerical convergence study for the test case (2) with $\eta = 1$ and $T = 0.5$. We compute the discrete L^2 error at the final time:

$$\|\mathbf{u}^N - \mathbf{u}(T)\|_{h,2} = \left(\sum_{i,j=0}^{M-1} |K_{i,j}| \, |u_{i,j}^N - u(T, \mathbf{x}_{i,j})|^2 \right)^{1/2},$$

where $\mathbf{u}^N = (u_{i,j}^N)$ is the numerical solution at time $T = N\Delta t$ and $\mathbf{u}(T) = (u(T, \mathbf{x}_{i,j}))$ is the exact solution evaluated at the cell centers.

We run the simulation for a series of meshes with $k = 1, 2, \ldots, 6$ (so $M = 2^k$ and $h = 1/M$). The time step Δt is chosen according to the CFL condition derived in Q6. The computed errors, estimated orders of convergence, and computation times (in seconds) are reported in Table 1.

k	M	h	Error L^2	Order	Time (s)
1	2	0.50	0.02487	0.00	0.00
2	4	0.25	0.01270	0.97	0.00
3	8	0.13	0.00671	0.92	0.00
4	16	0.06	0.00353	0.93	0.03
5	32	0.03	0.00182	0.95	0.34
6	64	0.02	0.00093	0.97	5.52

Table 1: Convergence results for the finite volume scheme (3)-(4) with $\eta = 1$, $T = 0.5$.

The order of convergence is computed as:

$$\text{Order} = \frac{\log(\text{Error}_k/\text{Error}_{k-1})}{\log(h_k/h_{k-1})}.$$

We observe that the error decreases as h decreases. The order of convergence stabilizes around 0.92 to 0.97, indicating approximately first-order convergence. This is consistent with the theoretical expectation for a first-order accurate scheme in space (using upwind for advection) combined with the forward Euler time stepping.

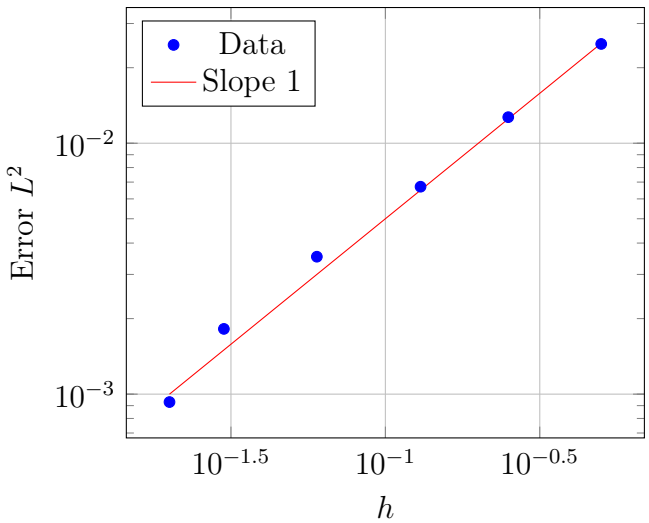

Figure 1: Log-log plot for Q7. The red line has slope 1.

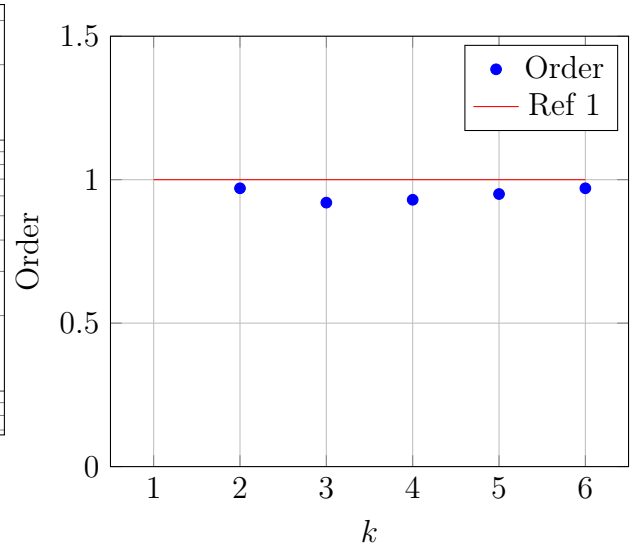

Figure 2: Order of convergence for Q7.

1.9 Q8: General Flux Framework and Equivalence with Upwind

We considered the function $B(z) = 1 + \max(-z, 0)$. Let check the properties (a)-(d):

- (a) $B(z) \geq 0$ for all $z \in \mathbb{R}$:
  If $z \leq 0$, then $\max(-z, 0) = -z \geq 0$, so $B(z) = 1 - z \geq 1 > 0$.
  If $z > 0$, then $\max(-z, 0) = 0$, so $B(z) = 1 > 0$.
  Hence $B(z) \geq 0$ for all z.

- (b) $B(0) = 1 + \max(-0, 0) = 1 + 0 = 1$.

- (c) B is continuous and piecewise $\mathcal{C}^1$:
  For $z < 0$, $B(z) = 1 - z$ (affine, hence $\mathcal{C}^1$).
  For $z > 0$, $B(z) = 1$ (constant, hence $\mathcal{C}^1$).
  At $z = 0$, we have $\lim_{z\to 0^-} B(z) = 1$ and $\lim_{z\to 0^+} B(z) = 1$, so B is continuous.
  The derivative for $z < 0$ is $B'(z) = -1$, and for $z > 0$ is $B'(z) = 0$. The left and right derivatives at 0 are different, but B is still piecewise $\mathcal{C}^1$.

- (d) $B(-z) - B(z) = z$ for all $z \in \mathbb{R}$:
  Compute:
  $$B(-z) = 1 + \max(z, 0), \quad B(z) = 1 + \max(-z, 0).$$
  Then
  $$B(-z) - B(z) = \max(z, 0) - \max(-z, 0).$$
  If $z \geq 0$, then $\max(z, 0) = z$ and $\max(-z, 0) = 0$, so $B(-z) - B(z) = z$.
  If $z < 0$, then $\max(z, 0) = 0$ and $\max(-z, 0) = -z$, so $B(-z) - B(z) = 0 - (-z) = z$.
  Thus the property holds.

Now, we show that with this B, the flux defined by (5) reduces to the upwind scheme (4) with the definitions from Q4.

Recall the flux (5):

$$\mathcal{F}^n_{K_{i,j},\sigma} = \begin{cases} \frac{|\sigma|}{d_\sigma}\big(B(-V_{K_{i,j},\sigma}d_\sigma)u^n_{i,j} - B(V_{K_{i,j},\sigma}d_\sigma)u^n_{i',j'}\big) & \text{for } \sigma = K_{i,j} \mid K_{i',j'} \in \mathcal{E}^{int}_h, \\ \frac{|\sigma|}{d_\sigma}\big(B(-V_{K_{i,j},\sigma}d_\sigma)u^n_{i,j} - B(V_{K_{i,j},\sigma}d_\sigma)u^D_\sigma\big) & \text{for } \sigma \in \mathcal{E}^D_h, \\ 0 & \text{for } \sigma \in \mathcal{E}^N_h. \end{cases}$$

Let $z = V_{K_{i,j},\sigma}d_\sigma$. Then:

$$B(-z) = 1 + \max(z, 0), \quad B(z) = 1 + \max(-z, 0).$$

For an interior edge, the flux becomes:

$$\mathcal{F}^n_{K_{i,j},\sigma} = \frac{|\sigma|}{d_\sigma}\big((1 + \max(z, 0))u^n_{i,j} - (1 + \max(-z, 0))u^n_{i',j'}\big).$$

We distinguish two cases:

- If $z \geq 0$ (i.e., $V_{K_{i,j},\sigma} \geq 0$):

  $$\max(z, 0) = z, \quad \max(-z, 0) = 0.$$

  Then

  $$\mathcal{F}^n_{K_{i,j},\sigma} = \frac{|\sigma|}{d_\sigma}\big((1 + z)u^n_{i,j} - 1 \cdot u^n_{i',j'}\big) = \frac{|\sigma|}{d_\sigma}(u^n_{i,j} - u^n_{i',j'}) + |\sigma|V_{K_{i,j},\sigma}u^n_{i,j}.$$

  This is exactly $\mathcal{F}^{L,n}_{K_{i,j},\sigma} + \mathcal{F}^{T,n}_{K_{i,j},\sigma}$ with the upwind choice: the diffusive flux is $-|\sigma|(u^n_{i',j'} - u^n_{i,j})/d_\sigma$ and the advective flux is $|\sigma|V_{K_{i,j},\sigma}u^n_{i,j}$.

- If $z < 0$ (i.e., $V_{K_{i,j},\sigma} < 0$):

  $$\max(z, 0) = 0, \quad \max(-z, 0) = -z.$$

  Then

  $$\mathcal{F}^n_{K_{i,j},\sigma} = \frac{|\sigma|}{d_\sigma}\big(1 \cdot u^n_{i,j} - (1 - z)u^n_{i',j'}\big) = \frac{|\sigma|}{d_\sigma}(u^n_{i,j} - u^n_{i',j'}) + |\sigma|V_{K_{i,j},\sigma}u^n_{i',j'}.$$

  Again, this matches the upwind scheme: diffusive flux as before and advective flux $|\sigma|V_{K_{i,j},\sigma}u^n_{i',j'}$.

For a Dirichlet edge, the same computation applies, replacing $u^n_{i',j'}$ with u^D_σ. For a Neumann edge, the flux is zero, which is consistent with the upwind scheme because on Neumann edges we impose zero total flux.

Thus, with $B(z) = 1 + \max(-z, 0)$, the flux (5) is equivalent to the upwind discretization (4).

1.10 Q9: Properties of the Bernoulli Function

Consider the Bernoulli function:

$$B(z) = \begin{cases} \dfrac{z}{e^z - 1}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

We verify the properties (a)-(d):

- (a) Positivity: $B(z) \geq 0$ for all $z \in \mathbb{R}$.
  For $z > 0$, $e^z - 1 > 0$, so $B(z) > 0$.
  For $z < 0$, write $B(z) = \dfrac{ze^{-z}}{1 - e^{-z}}$. Since $z < 0$, $ze^{-z} < 0$, and $1 - e^{-z} < 0$, so the ratio is positive.
  At $z = 0$, $B(0) = 1 > 0$.

- (b) $B(0) = 1$ by definition.

- (c) Continuity and piecewise $\mathcal{C}^1$:
  The function is smooth ($\mathcal{C}^\infty$) for $z \neq 0$. We need to check continuity at 0. Using the expansion $e^z = 1 + z + \frac{z^2}{2} + \cdots$, we have for $z \neq 0$:

  $$B(z) = \frac{z}{e^z - 1} = \frac{1}{1 + \frac{z}{2} + \cdots} \to 1 \quad \text{as } z \to 0.$$

  Hence B is continuous at 0. Its derivative exists for all z (including 0 via limit), and B is actually $\mathcal{C}^\infty$ on $\mathbb{R}$.

- (d) The identity $B(-z) - B(z) = z$.
  Compute:

  $$B(-z) = \frac{-z}{e^{-z} - 1} = \frac{ze^z}{e^z - 1}.$$

  Then

  $$B(-z) - B(z) = \frac{ze^z}{e^z - 1} - \frac{z}{e^z - 1} = z\frac{e^z - 1}{e^z - 1} = z.$$

  This holds for all $z \neq 0$. At $z = 0$, both sides are 0.

Therefore, the Bernoulli function satisfies all the assumptions (a)-(d).

1.11 Q10-11: Formulation and Stability of the : Scharfetter-Gummel Scheme

Write the Scharfetter–Gummel scheme (3)–(5)–(6) in the form $u_{i,j}^{n+1} = L_{i,j}\left((u_{i',j'}^n), (u_\sigma^D)\right) + \Delta t f_{i,j}$, where $L_{i,j}$ is linear, and give a condition on Δt ensuring that $L_{i,j}$ is a convex combination.

Starting from the scheme:

$$\frac{|K_{i,j}|}{\Delta t}(u_{i,j}^{n+1} - u_{i,j}^{n}) + \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^{n} = |K_{i,j}| f_{i,j},$$

with the flux given by (5) and the Bernoulli function (6), we isolate $u_{i,j}^{n+1}$:

$$u_{i,j}^{n+1} = u_{i,j}^{n} - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^{n} + \Delta t f_{i,j}.$$

Substituting the flux expressions and grouping the terms, we define the following coefficients:

- Coefficient of $u_{i,j}^{n}$:

$$\alpha_{i,j} = 1 - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \frac{|\sigma|}{d_\sigma} B\big(-V_{K_{i,j},\sigma}\, d_\sigma\big).$$

- For each interior edge $\sigma = K_{i,j} \mid K_{i',j'} \in \mathcal{E}_h^{int}$:

$$\beta_\sigma = \frac{\Delta t}{|K_{i,j}|} \frac{|\sigma|}{d_\sigma} B\big(V_{K_{i,j},\sigma}\, d_\sigma\big).$$

- For each Dirichlet edge $\sigma \in \mathcal{E}_h^D \cap \mathcal{E}_{K_{i,j}}$:

$$\gamma_\sigma = \frac{\Delta t}{|K_{i,j}|} \frac{|\sigma|}{d_\sigma} B\big(V_{K_{i,j},\sigma}\, d_\sigma\big).$$

- For Neumann edges, the flux is zero, so they contribute neither to $\alpha_{i,j}$ nor to the other coefficients (they are included in the sum for $\alpha_{i,j}$ but with $V_{K_{i,j},\sigma} = 0$ and the appropriate d_σ).

Then the scheme can be written as:

$$u_{i,j}^{n+1} = \alpha_{i,j}\, u_{i,j}^{n} + \sum_{\sigma \in \mathcal{E}_h^{int} \cap \mathcal{E}_{K_{i,j}}} \beta_\sigma\, u_{i',j'}^{n} + \sum_{\sigma \in \mathcal{E}_h^{D} \cap \mathcal{E}_{K_{i,j}}} \gamma_\sigma\, u_\sigma^{D} + \Delta t f_{i,j}.$$

Thus, $u_{i,j}^{n+1} = L_{i,j}\left((u_{i',j'}^{n}), (u_\sigma^{D})\right) + \Delta t f_{i,j}$ with $L_{i,j}$ a linear function.

For $L_{i,j}$ to be a convex combination (when $f = 0$), all coefficients must be nonnegative and their sum must equal 1. The sum is:

$$\alpha_{i,j} + \sum_{\text{int}} \beta_\sigma + \sum_{\text{D}} \gamma_\sigma = 1,$$

which holds by construction (it follows from the conservation property of the fluxes). The nonnegativity conditions are:

- $\alpha_{i,j} \geq 0$:

$$1 - \frac{\Delta t}{|K_{i,j}|} \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \frac{|\sigma|}{d_\sigma} B(-V_{K,\sigma} d_\sigma) \geq 0 \quad \Longrightarrow \quad \Delta t \leq \frac{|K_{i,j}|}{\displaystyle\sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \frac{|\sigma|}{d_\sigma} B(-V_{K,\sigma} d_\sigma)}.$$

- $\beta_\sigma \geq 0$ and $\gamma_\sigma \geq 0$: these are automatically satisfied because $B(z) \geq 0$ for all z (property (a)) and all other factors are positive.

Hence, the critical condition is the one on $\alpha_{i,j}$. To obtain an explicit bound in terms of h and $\|\mathbf{V}\|_{L^\infty(\Omega)}$, we estimate the denominator. For a uniform Cartesian mesh, $|K_{i,j}| = h^2$, $|\sigma| = h$, and $d_\sigma = h$ (interior edges) or $h/2$ (boundary edges). Using the inequality $B(z) \leq 1 + |z|$ for all $z \in \mathbb{R}$, we have:

$$B(-V_{K,\sigma} d_\sigma) \leq 1 + |V_{K,\sigma}| d_\sigma.$$

Therefore,

$$\sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \frac{|\sigma|}{d_\sigma} B(-V_{K,\sigma} d_\sigma) \leq \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \frac{|\sigma|}{d_\sigma} + \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| |V_{K,\sigma}|.$$

The first sum is bounded by 4 for an interior cell (and at most 6 for a boundary cell). The second sum involves only the vertical edges (two per cell), and $|V_{K,\sigma}| \leq \|\mathbf{V}\|_{L^\infty(\Omega)} = |\eta|$. Hence,

$$\sum_{\sigma \in \mathcal{E}_{K_{i,j}}} |\sigma| |V_{K,\sigma}| \leq 2h \|\mathbf{V}\|_{L^\infty(\Omega)}.$$

Taking the worst case (interior cell, with the bound 4 for the first sum), a sufficient condition for all cells is:

$$\Delta t \leq \frac{h^2}{4 + 2h \|\mathbf{V}\|_{L^\infty(\Omega)}}.$$

If this condition is satisfied and $f \equiv 0$, then all coefficients $\alpha_{i,j}, \beta_\sigma, \gamma_\sigma$ are nonnegative and sum to 1, so $L_{i,j}$ is a convex combination. Consequently, the scheme satisfies a discrete maximum principle and is stable in the maximum norm.

1.12 Q12: Numerical Convergence Study of the Scharfetter-Gummel Scheme

We implement the Scharfetter–Gummel scheme (3)–(5)–(6) and perform a convergence study for the test case (2) with $\eta = 1$ and $T = 0.5$. The simulations are run on a sequence of uniformly refined Cartesian meshes with $k = 1, \ldots, 6$.

We observe that the error decreases much more rapidly than in the upwind case. The estimated orders are consistently close to 2 (stabilizing at 2.02 for $k = 6$), indicating second-order convergence in space for the SG scheme.

The computation times increase considerably (reaching 29.94s for $k = 6$) due to the stricter CFL condition ($\Delta t \propto h^2$) and the increased number of cells. However, the SG scheme achieves significantly higher accuracy for the same mesh size compared to the upwind scheme.

k	M	h	Error L^2	Order	Time (s)
1	2	0.50	0.00483	0.00	0.00
2	4	0.25	0.00079	2.60	0.00
3	8	0.13	0.00017	2.20	0.01
4	16	0.06	0.00004	2.08	0.12
5	32	0.03	0.00001	2.04	1.89
6	64	0.02	0.00000	2.02	29.94

Table 2: Convergence results for the Scharfetter–Gummel scheme ($\eta = 1$, $T = 0.5$).

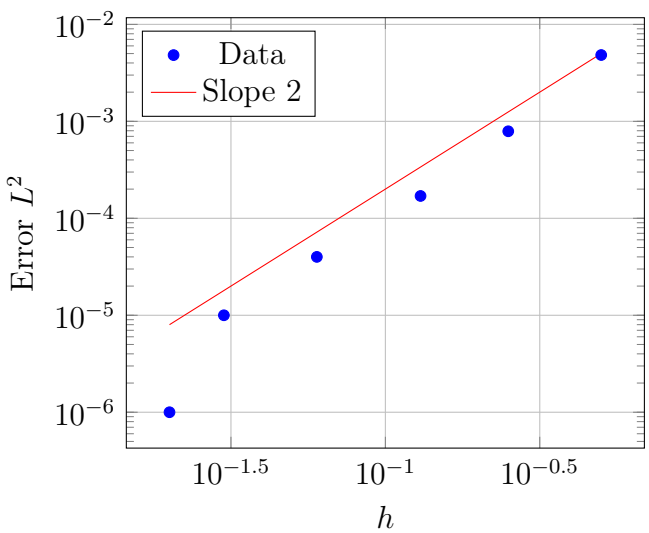

Figure 3: Log-log plot for SG scheme. Slope 2 for reference.

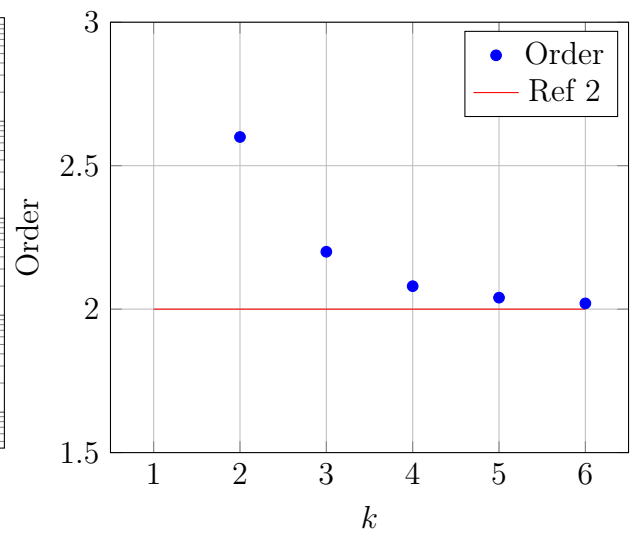

Figure 4: Order of convergence for SG scheme.

1.13 Question 13: Long-time Behavior and Steady-state Convergence

1.13.1 a) Analytical Steady-state Solution

The time-dependent analytical solution to the advection-diffusion equation is given by:

$$u(t,x) = e^{\eta x} + e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} e^{\frac{\eta}{2}x} \sin(\pi x) \tag{1}$$

As $t \to \infty$, the transient term $e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t}$ decays exponentially toward zero. Consequently, the theoretical steady-state solution $u^\infty(x)$ is defined as:

$$u^\infty(x) = e^{\eta x} \tag{2}$$

1.13.2 b) Numerical Simulation and Discrete Steady-state

We performed simulations for $\eta = 1$ using both the **Upwind** and **Scharfetter-Gummel (SG)** schemes. The time loop was terminated when the Cauchy residual $\|u^{n+1} - u^n\|_{h,2}$ fell

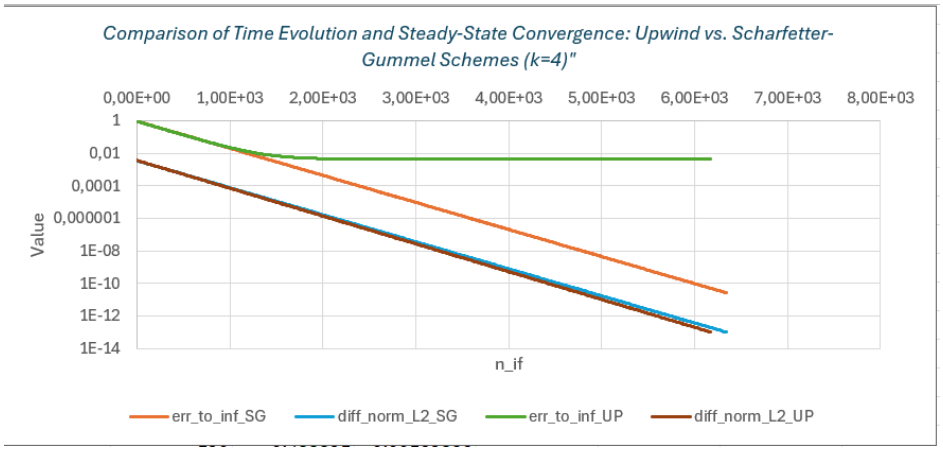

Figure 5: Comparison of Time Evolution and Steady-State Convergence: Upwind vs. Scharfetter-Gummel Schemes (k=4)

below the tolerance of 10^{-13}. The final state reached at this tolerance is denoted as the discrete steady-state u^{n_1}.

The quantitative results for different mesh levels k are summarized in Table 1.13.2:

Scheme	k	M	h	**Error** L^2 $(u^{n_1} - u^\infty)$	**Order**	n_1 (Iter)
SG	2	4	0.25	1.696×10^{-12}	-	493
	4	16	0.0625	2.601×10^{-11}	-1.96	6,344
	6	64	0.0156	4.078×10^{-10}	-1.99	87,840
Upwind	2	4	0.25	1.418×10^{-2}	-	454
	4	16	0.0625	4.275×10^{-3}	0.89	6,179
	6	64	0.0156	1.143×10^{-3}	0.97	87,227

Table 3: Steady-state convergence results for SG and Upwind schemes.

1.13.3 c) Time Evolution Analysis and Graphical Observations

The time evolution for $k = 4$ $(M = 16, h = 0.0625)$ was analyzed by plotting the theoretical error $\|u^n - u^\infty\|_{h,2}$ and the Cauchy residual $\|u^n - u^{n_1}\|_{h,2}$ on a semilog-y scale.

1.13.4 Observations and Physical Interpretation

- **Exponential Decay:** For both schemes, the Cauchy residual decreases linearly on a logarithmic scale, confirming that the system converges exponentially toward the equilibrium u^{n_1}.
- **SG Scheme Accuracy:** The SG scheme achieves an error level of 2.60×10^{-11} for $k = 4$. This confirms that the SG flux is analytically exact for the exponential steady-state $e^{\eta x}$.

- **Upwind Scheme Limitation:** The Upwind scheme levels off at a plateau of 4.27×10^{-3}. This represents the irreducible spatial discretization error associated with the first-order scheme.

- **Numerical Noise and Negative Orders:** In the SG results, we observe negative convergence orders (e.g., -1.99). This is a numerical artifact: the error has reached the machine precision floor. As the mesh is refined (higher M), the increased number of iterations (from 493 to 87,840) leads to a slight accumulation of floating-point rounding errors, which the order calculation misinterprets as a loss of convergence.

- **Efficiency vs. Quality:** The Upwind scheme is computationally faster (52.79s vs 339.58s for $k = 6$) and requires slightly fewer iterations to stabilize. This is due to its lower computational complexity and higher numerical diffusion, which dampens transients faster, albeit toward a significantly less accurate result.

1.13.5 Conclusion

Reducing h systematically improves the Upwind steady-state precision at a first-order rate. For the SG scheme, the solution is already near-perfect at coarse mailling ($k = 2$), proving its absolute superiority for capturing stationary advection-diffusion profiles.

2 Part II : Finite Element Approximation

2.1 Q14. Semi-variational formulation

We consider the time-dependent advection–diffusion problem (1) from the project:

$$\begin{cases} \partial_t u + \operatorname{div}(-\nabla u + \mathbf{V}u) = f & \text{in } (0,T] \times \Omega, \\ u = u_D & \text{on } (0,T] \times \Gamma_D, \\ (-\nabla u + \mathbf{V}u) \cdot \mathbf{n} = 0 & \text{on } (0,T] \times \Gamma_N, \\ u(0,\cdot) = u_0 & \text{in } \Omega, \end{cases}$$

where $\mathbf{n}$ is the outward unit normal on $\partial\Omega$, and $\mathbf{V}$ satisfies $\operatorname{div}(\mathbf{V}) = 0$ in Ω and $\mathbf{V} \cdot \mathbf{n} = 0$ on Γ_N.

The variational framework requires the solution to belong to $H^1(\Omega)$ for almost every t. The Dirichlet condition is imposed strongly via the space

$$H_0^{1,D}(\Omega) := \{v \in H^1(\Omega) \mid \gamma(v)|_{\Gamma_D} = 0\},$$

where $\gamma : H^1(\Omega) \to H^{1/2}(\partial\Omega)$ is the trace operator. Since Γ_D has positive measure, the Poincaré inequality holds: there exists $C_P > 0$ such that

$$\|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_0^{1,D}(\Omega).$$

Hence $\|\nabla \cdot\|_{L^2(\Omega)}$ is a norm equivalent to the H^1-norm on $H_0^{1,D}(\Omega)$, which is therefore a Hilbert space.

Given $u_D \in H^{1/2}(\Gamma_D)$, we admit the existence of a lifting $\tilde{u}^D \in H^1(\Omega)$ satisfying

$$\gamma(\tilde{u}^D)|_{\Gamma_D} = u_D, \qquad \|\tilde{u}^D\|_{H^1(\Omega)} \leq C_{L,D} \|u_D\|_{H^{1/2}(\Gamma_D)}.$$

We decompose the unknown as

$$u(t) = \tilde{u}(t) + \tilde{u}^D, \quad \text{with } \tilde{u}(t) \in H_0^{1,D}(\Omega).$$

Let $v \in H_0^{1,D}(\Omega)$. Multiplying the equation by v and integrating over Ω yields

$$\int_\Omega \partial_t u \, v \, dx + \int_\Omega \operatorname{div}(-\nabla u + \mathbf{V}u) \, v \, dx = \int_\Omega f v \, dx.$$

Applying Green's formula to the divergence term gives

$$\int_\Omega \operatorname{div}(-\nabla u + \mathbf{V}u) v = -\int_\Omega (-\nabla u + \mathbf{V}u) \cdot \nabla v \, dx + \int_{\partial\Omega} (-\nabla u + \mathbf{V}u) \cdot \mathbf{n} \, v \, ds.$$

The boundary integral vanishes because $v = 0$ on Γ_D and $(-\nabla u + \mathbf{V}u) \cdot \mathbf{n} = 0$ on Γ_N. Hence we obtain

$$\int_\Omega \partial_t u \, v + \int_\Omega \nabla u \cdot \nabla v - \int_\Omega (\mathbf{V}u) \cdot \nabla v = \int_\Omega f v.$$

Defining the bilinear form

$$a(w,v) := \int_\Omega \nabla w \cdot \nabla v \, dx - \int_\Omega (\mathbf{V} w) \cdot \nabla v \, dx,$$

and substituting $u = \tilde{u} + \tilde{u}^D$, we arrive at the semi-variational formulation:
Find $\tilde{u} : (0,T) \to H_0^{1,D}(\Omega)$ such that for almost every $t \in (0,T)$,

$$\frac{d}{dt} \int_\Omega \tilde{u}(t) v + a(\tilde{u}(t), v) = \int_\Omega f v - a(\tilde{u}^D, v) \quad \forall v \in H_0^{1,D}(\Omega),$$

with the initial condition $\tilde{u}(0) = u_0 - \tilde{u}^D$ in $L^2(\Omega)$. This is exactly formulation (7) of the project.

2.2 Q15. Well-posedness of the steady problem

The steady problem associated with (1) reads: find $u^\infty = \tilde{u}^\infty + \tilde{u}^D$, with $\tilde{u}^\infty \in H_0^{1,D}(\Omega)$, such that

$$a(\tilde{u}^\infty, v) = \int_\Omega f v - a(\tilde{u}^D, v) \quad \forall v \in H_0^{1,D}(\Omega). \tag{8}$$

1. Continuity of a. For $w, v \in H_0^{1,D}(\Omega)$,

$$|a(w,v)| \le \|\nabla w\|_{L^2} \|\nabla v\|_{L^2} + \|\mathbf{V}\|_{L^\infty} \|w\|_{L^2} \|\nabla v\|_{L^2}.$$

By the Poincaré inequality, $\|w\|_{L^2} \le C_P \|\nabla w\|_{L^2}$, hence

$$|a(w,v)| \le (1 + C_P \|\mathbf{V}\|_{L^\infty}) \|\nabla w\|_{L^2} \|\nabla v\|_{L^2}.$$

Thus a is continuous on $H_0^{1,D}(\Omega)$ endowed with the norm $\|\nabla \cdot\|_{L^2}$.

2. Coercivity of a. For $v \in H_0^{1,D}(\Omega)$,

$$a(v,v) = \|\nabla v\|_{L^2}^2 - \int_\Omega (\mathbf{V} v) \cdot \nabla v.$$

Using $\operatorname{div}(\mathbf{V}) = 0$ and $\mathbf{V} \cdot \mathbf{n} = 0$ on Γ_N, integration by parts gives

$$\int_\Omega (\mathbf{V} v) \cdot \nabla v = \frac{1}{2} \int_\Omega \mathbf{V} \cdot \nabla (v^2) = 0.$$

Therefore

$$a(v,v) = \|\nabla v\|_{L^2}^2,$$

which shows that a is coercive with constant $\alpha = 1$.

3. Continuity of the linear form. Define $L(v) := \int_\Omega f v - a(\tilde{u}^D, v)$. Then

$$|L(v)| \le \|f\|_{L^2} \|v\|_{L^2} + |a(\tilde{u}^D, v)| \le C_P \|f\|_{L^2} \|\nabla v\|_{L^2} + M \|\tilde{u}^D\|_{H^1} \|\nabla v\|_{L^2},$$

where $M = 1 + C_P \|\mathbf{V}\|_{L^\infty}$. Hence L is continuous on $H_0^{1,D}(\Omega)$.

4. Application of Lax–Milgram. Since $(H_0^{1,D}(\Omega), \|\nabla \cdot\|_{L^2})$ is a Hilbert space, a is continuous and coercive, and L is continuous, the Lax–Milgram theorem guarantees existence and uniqueness of $\tilde{u}^\infty \in H_0^{1,D}(\Omega)$ solving (8). Consequently, the steady problem is well-posed in the sense of Hadamard.

2.3 Q16. Long-time behaviour

Let $u(t)$ be the solution of (7) and u^∞ the solution of (8). Both are written with the same lifting:

$$u(t) = \tilde{u}(t) + \tilde{u}^D, \qquad u^\infty = \tilde{u}^\infty + \tilde{u}^D, \quad \tilde{u}(t), \tilde{u}^\infty \in H_0^{1,D}(\Omega).$$

Set $w(t) := u(t) - u^\infty = \tilde{u}(t) - \tilde{u}^\infty \in H_0^{1,D}(\Omega)$. Subtracting (8) from (7) gives, for all $v \in H_0^{1,D}(\Omega)$,

$$\frac{d}{dt}\int_\Omega w(t)v + a(w(t), v) = 0. \tag{A}$$

Take $v = w(t)$ (admissible because $w(t) \in H_0^{1,D}(\Omega)$). Using $a(w, w) = \|\nabla w\|_{L^2}^2$ (coercivity) we obtain

$$\frac{1}{2}\frac{d}{dt}\|w(t)\|_{L^2}^2 + \|\nabla w(t)\|_{L^2}^2 = 0.$$

The Poincaré inequality $\|w\|_{L^2} \le C_P \|\nabla w\|_{L^2}$ yields

$$\frac{d}{dt}\|w(t)\|_{L^2}^2 + \frac{2}{C_P^2}\|w(t)\|_{L^2}^2 \le 0.$$

Defining $y(t) := \|w(t)\|_{L^2}^2$, this is a differential inequality $y'(t) + 2\beta y(t) \le 0$ with $\beta = C_P^{-2}$. Grönwall's lemma gives

$$y(t) \le y(0)e^{-2\beta t},$$

hence

$$\|w(t)\|_{L^2} \le e^{-\beta t}\|w(0)\|_{L^2}.$$

Recalling $w(t) = u(t) - u^\infty$ and $w(0) = u_0 - u^\infty$, we finally obtain

$$\boxed{\|u(t) - u^\infty\|_{L^2(\Omega)} \le e^{-\beta t}\|u_0 - u^\infty\|_{L^2(\Omega)}}, \qquad \beta = \frac{1}{C_P^2} > 0.$$

Thus the solution converges exponentially to the steady state in the L^2-norm, with a rate β that depends only on the Poincaré constant of the domain.

2.4 Q17. Céa-type estimate

Let $U_h \subset H^1(\Omega) \cap C^0(\overline{\Omega})$ be a conforming finite-element space and define

$$U_{h,0}^D := U_h \cap H_0^{1,D}(\Omega).$$

The discrete steady problem (9) seeks $u_h^\infty = \tilde{u}_h^\infty + \tilde{u}^D$, with $\tilde{u}_h^\infty \in U_{h,0}^D$, such that

$$a(\tilde{u}_h^\infty, v_h) = \int_\Omega f v_h - a(\tilde{u}^D, v_h) \quad \forall v_h \in U_{h,0}^D.$$

Let $e := u^\infty - u_h^\infty = \tilde{u}^\infty - \tilde{u}_h^\infty \in H_0^{1,D}(\Omega)$. For any $v_h \in U_h$ with $\gamma(v_h)|_{\Gamma_D} = u^D$, we have $v_h - u_h^\infty \in U_{h,0}^D$. Using Galerkin orthogonality and the coercivity plus continuity of a, one obtains

$$\|\nabla e\|_{L^2} \leq \|\nabla(u^\infty - v_h)\|_{L^2} + \|\mathbf{V}\|_{L^\infty}\|u^\infty - v_h\|_{L^2}.$$

Taking the infimum over all such v_h yields

$$\boxed{\|\nabla(u^\infty - u_h^\infty)\|_{L^2(\Omega)} \leq \inf_{\substack{v_h \in U_h \\ \gamma(v_h)|_{\Gamma_D} = u^D}} \big(\|\nabla(u^\infty - v_h)\|_{L^2} + \|\mathbf{V}\|_{L^\infty}\|u^\infty - v_h\|_{L^2}\big)}. \tag{10}$$

2.5 Q18. A priori error estimate and the Péclet number

Assume $u^\infty \in H^2(\Omega)$ and let $\mathcal{I}_h : H^1(\Omega) \cap C^0(\overline{\Omega}) \to U_h$ be an interpolation operator satisfying

$$\|\nabla(v - \mathcal{I}_h v)\|_{L^2} \leq Ch|v|_{H^2}, \quad \|v - \mathcal{I}_h v\|_{L^2} \leq C'h^2|v|_{H^2} \quad \forall v \in H^2(\Omega),$$

with C, C' independent of h, and preserving the Dirichlet condition: $\mathcal{I}_h(H_0^{1,D}(\Omega) \cap C^0(\overline{\Omega})) \subseteq U_{h,0}^D$.

Choosing $v_h = \mathcal{I}_h u^\infty$ in (10) gives

$$\|\nabla(u^\infty - u_h^\infty)\|_{L^2} \leq Ch|u^\infty|_{H^2} + \|\mathbf{V}\|_{L^\infty} C'h^2|u^\infty|_{H^2}.$$

Introducing the Péclet number $\mathrm{Pe} := h\|\mathbf{V}\|_{L^\infty(\Omega)}$, we obtain

$$\boxed{\|\nabla(u^\infty - u_h^\infty)\|_{L^2(\Omega)} \leq h|u^\infty|_{H^2(\Omega)}\big(C + C'\mathrm{Pe}\big)}.$$

Discussion.

- If $\mathrm{Pe} \ll 1$ (diffusion-dominated), the bound is $O(h)$, the optimal rate for $\mathbb{Q}^1$ elements.
- If $\mathrm{Pe} \gg 1$ (advection-dominated), the term $C'\mathrm{Pe}$ dominates and the error may be large unless h is small enough that $h\|\mathbf{V}\|_{L^\infty} \lesssim 1$. This illustrates the need for stabilised methods at high Péclet numbers.

2.6 Question 19: Unisolvance of the $\mathbb{Q}^1$ finite element

Let us prove that the triple $(K, \mathbb{Q}^1(K), \{\tau_\ell\}_{0 \leq \ell \leq 3})$, with $\tau_\ell : \mathbb{Q}^1(K) \to \mathbb{R}$ such that $\tau_\ell(v) := v(\boldsymbol{\nu}_\ell)$ for all $v \in \mathbb{Q}^1(K)$ and $0 \leq \ell \leq 3$, is **unisolvent** as soon as the square K is non-degenerate.

Proof Let K be a non-degenerate square. The space $\mathbb{Q}^1(K)$ is defined as:

$$\mathbb{Q}^1(K) = \{\alpha + \beta_1 x + \beta_2 y + \gamma xy \mid (\alpha, \beta_1, \beta_2, \gamma) \in \mathbb{R}^4\}.$$

Thus, $\dim \mathbb{Q}^1(K) = 4$.

The set of degrees of freedom $\Sigma = \{\tau_0, \tau_1, \tau_2, \tau_3\}$ has exactly 4 elements, each defined by evaluation at a vertex $\boldsymbol{\nu}_\ell$ of K:

$$\tau_\ell(v) = v(\boldsymbol{\nu}_\ell), \quad \ell = 0, 1, 2, 3.$$

To prove unisolvence, it suffices to show that if $v \in \mathbb{Q}^1(K)$ satisfies $\tau_\ell(v) = 0$ for all ℓ, then $v \equiv 0$.

Since K is a non-degenerate square, we can assume (via an affine change of coordinates) that K is the unit square $[0, 1]^2$ with vertices:

$$\boldsymbol{\nu}_0 = (0, 0), \quad \boldsymbol{\nu}_1 = (1, 0), \quad \boldsymbol{\nu}_2 = (1, 1), \quad \boldsymbol{\nu}_3 = (0, 1).$$

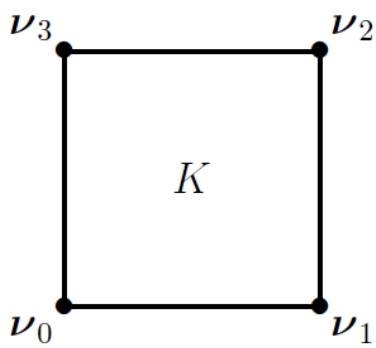

Figure 6: cell K

Let $v(x, y) = \alpha + \beta_1 x + \beta_2 y + \gamma xy$. The conditions $v(\boldsymbol{\nu}_\ell) = 0$ yield the linear system:

$$\begin{aligned}
&v(0,0) = \alpha = 0, \\
&v(1,0) = \alpha + \beta_1 = 0 \quad \Rightarrow \beta_1 = 0, \\
&v(0,1) = \alpha + \beta_2 = 0 \quad \Rightarrow \beta_2 = 0, \\
&v(1,1) = \alpha + \beta_1 + \beta_2 + \gamma = 0 \quad \Rightarrow \gamma = 0.
\end{aligned}$$

Hence, $\alpha = \beta_1 = \beta_2 = \gamma = 0$, so $v \equiv 0$.

Therefore, the mapping $\Phi : \mathbb{Q}^1(K) \to \mathbb{R}^4$ defined by $\Phi(v) = (v(\boldsymbol{\nu}_0), v(\boldsymbol{\nu}_1), v(\boldsymbol{\nu}_2), v(\boldsymbol{\nu}_3))$ is injective. Since $\dim \mathbb{Q}^1(K) = 4 = \dim \mathbb{R}^4$, Φ is an isomorphism. Consequently, the triple $(K, \mathbb{Q}^1(K), \{\tau_\ell\}_{\ell=0}^3)$ is unisolvent. $\square$

2.7 Question 20: Local shape functions of the $\mathbb{Q}^1$ finite element

Let us verify that, for a given square $K \in \mathcal{T}_h$, the local shape functions for the $\mathbb{Q}^1$-FE are given, for $0 \le \ell \le 3$, by:

$$\varphi_\ell(\boldsymbol{x}) := \frac{\big(x - \nu_{(\ell+2)[4],x}\big)\big(y - \nu_{(\ell+2)[4],y}\big)}{\big(\nu_{\ell,x} - \nu_{(\ell+2)[4],x}\big)\big(\nu_{\ell,y} - \nu_{(\ell+2)[4],y}\big)} \qquad \forall \boldsymbol{x} = (x, y) \in \overline{K},$$

where $(\ell + 2)[4]$ denotes $(\ell + 2) \mod 4$.

Verification We consider a non-degenerate square K. By an affine change of coordinates, we can assume without loss of generality that K is the unit square $[0,1]^2$, with vertices labelled as:

$$\boldsymbol{\nu}_0 = (0,0), \quad \boldsymbol{\nu}_1 = (1,0), \quad \boldsymbol{\nu}_2 = (1,1), \quad \boldsymbol{\nu}_3 = (0,1).$$

Let us verify the formula for each $\ell = 0, 1, 2, 3$.

Case $\ell = 0$ We have $(\ell+2)[4] = 2$, so $\boldsymbol{\nu}_{(\ell+2)[4]} = \boldsymbol{\nu}_2 = (1,1)$. Then:

$$\varphi_0(x,y) = \frac{(x-1)(y-1)}{(0-1)(0-1)} = (x-1)(y-1).$$

Evaluation at vertices:

$$\begin{aligned}
\varphi_0(\boldsymbol{\nu}_0) &= (0-1)(0-1) = 1, \\
\varphi_0(\boldsymbol{\nu}_1) &= (1-1)(0-1) = 0, \\
\varphi_0(\boldsymbol{\nu}_2) &= (1-1)(1-1) = 0, \\
\varphi_0(\boldsymbol{\nu}_3) &= (0-1)(1-1) = 0.
\end{aligned}$$

Case $\ell = 1$ We have $(\ell+2)[4] = 3$, so $\boldsymbol{\nu}_{(\ell+2)[4]} = \boldsymbol{\nu}_3 = (0,1)$. Then:

$$\varphi_1(x,y) = \frac{(x-0)(y-1)}{(1-0)(0-1)} = -x(y-1) = x(1-y).$$

Evaluation at vertices:

$$\begin{aligned}
\varphi_1(\boldsymbol{\nu}_0) &= 0 \cdot (1-0) = 0, \\
\varphi_1(\boldsymbol{\nu}_1) &= 1 \cdot (1-0) = 1, \\
\varphi_1(\boldsymbol{\nu}_2) &= 1 \cdot (1-1) = 0, \\
\varphi_1(\boldsymbol{\nu}_3) &= 0 \cdot (1-1) = 0.
\end{aligned}$$

Case $\ell = 2$ We have $(\ell+2)[4] = 0$, so $\boldsymbol{\nu}_{(\ell+2)[4]} = \boldsymbol{\nu}_0 = (0,0)$. Then:

$$\varphi_2(x,y) = \frac{(x-0)(y-0)}{(1-0)(1-0)} = xy.$$

Evaluation at vertices:

$$\begin{aligned}
\varphi_2(\boldsymbol{\nu}_0) &= 0 \cdot 0 = 0, \\
\varphi_2(\boldsymbol{\nu}_1) &= 1 \cdot 0 = 0, \\
\varphi_2(\boldsymbol{\nu}_2) &= 1 \cdot 1 = 1, \\
\varphi_2(\boldsymbol{\nu}_3) &= 0 \cdot 1 = 0.
\end{aligned}$$

Case $\ell = 3$ We have $(\ell+2)[4] = 1$, so $\boldsymbol{\nu}_{(\ell+2)[4]} = \boldsymbol{\nu}_1 = (1,0)$. Then:

$$\varphi_3(x,y) = \frac{(x-1)(y-0)}{(0-1)(1-0)} = -(x-1)y = (1-x)y.$$

Evaluation at vertices:

$$\begin{aligned}
\varphi_3(\boldsymbol{\nu}_0) &= (1-0)\cdot 0 = 0,\\
\varphi_3(\boldsymbol{\nu}_1) &= (1-1)\cdot 0 = 0,\\
\varphi_3(\boldsymbol{\nu}_2) &= (1-1)\cdot 1 = 0,\\
\varphi_3(\boldsymbol{\nu}_3) &= (1-0)\cdot 1 = 1.
\end{aligned}$$

In all cases, we have $\varphi_\ell(\boldsymbol{\nu}_m) = \delta_{\ell m}$.

Conclusion The given functions $\{\varphi_\ell\}_{\ell=0}^3$ are indeed the local shape functions of the $\mathbb{Q}^1$ finite element, as they:

1. Belong to $\mathbb{Q}^1(K)$ (each is a product of two linear factors, hence a bilinear polynomial),
2. Satisfy the Kronecker condition $\varphi_\ell(\boldsymbol{\nu}_m) = \delta_{\ell m}$ for all ℓ, m.

2.8 Question 21: Continuity of $\mathbb{Q}^1$ functions across edges

I this section we will Justify that a function v defined on $\overline{K} \cup \overline{L}$, with $K, L \in \mathcal{T}_h$ sharing the edge $\sigma := (\boldsymbol{\nu}_\flat, \boldsymbol{\nu}_\sharp) \in \mathcal{E}_h^{int}$, such that $v_{|K} \in \mathbb{Q}^1(K)$, $v_{|L} \in \mathbb{Q}^1(L)$, and $v_{|K}(\boldsymbol{\nu}_\flat) = v_{|L}(\boldsymbol{\nu}_\flat)$, $v_{|K}(\boldsymbol{\nu}_\sharp) = v_{|L}(\boldsymbol{\nu}_\sharp)$, satisfies $v \in C^0(\overline{K} \cup \overline{L})$.

[Continuity of piecewise polynomials matching at nodes] Let K and L be two adjacent cells sharing an edge σ. Let $v_K \in \mathbb{Q}^1(K)$ and $v_L \in \mathbb{Q}^1(L)$. If v_K and v_L coincide at the two vertices of σ, then $v_K = v_L$ on the whole edge σ.

Proof Consider the edge $\sigma = (\boldsymbol{\nu}_\flat, \boldsymbol{\nu}_\sharp)$. Without loss of generality, we can assume Indeed, on σ, $y = 0$ and:

$$v_K(x,0) = \alpha_K + \beta_K x + \gamma_K \cdot 0 + \delta_K x \cdot 0 = \alpha_K + \beta_K x.$$

Similarly, $v_L(x,0) = \alpha_L + \beta_L x$.

The conditions given are:

$$v_K(\boldsymbol{\nu}_\flat) = v_L(\boldsymbol{\nu}_\flat), \quad v_K(\boldsymbol{\nu}_\sharp) = v_L(\boldsymbol{\nu}_\sharp).$$

In our local coordinates, $\boldsymbol{\nu}_\flat = (0,0)$ and $\boldsymbol{\nu}_\sharp = (1,0)$. Thus:

$$v_K(0,0) = v_L(0,0) \quad \Rightarrow \quad \alpha_K = \alpha_L,$$

$$v_K(1,0) = v_L(1,0) \quad \Rightarrow \quad \alpha_K + \beta_K = \alpha_L + \beta_L \quad \Rightarrow \quad \beta_K = \beta_L.$$

Hence, $v_K(x,0) = v_L(x,0)$ for all $x \in [0,1]$. Therefore, v_K and v_L coincide on the entire edge σ.

Since v is defined as $v|_K = v_K$ and $v|_L = v_L$, and they match on σ, the function v is continuous across σ. As the only possible discontinuity in $\overline{K} \cup \overline{L}$ is across σ, we conclude that $v \in C^0(\overline{K} \cup \overline{L})$. □

This argument relies on the fact that a linear polynomial in one variable is uniquely determined by its values at two distinct points. This is a direct consequence of the unisolvence of the $\mathbb{P}^1$ element on a segment, which is inherited by the $\mathbb{Q}^1$ element when restricted to an edge.

2.9 Question 22: Existence of a discrete lifting of the Dirichlet datum

Justify that, for test-case (2), it is indeed possible to construct a lifting $\tilde{u}^D \in U_h$ of the Dirichlet datum u^D. In practice, this lifting will be taken such that $\tilde{u}^D(\boldsymbol{\nu}) = 0$ for all vertices $\boldsymbol{\nu}$ not pertaining to $\overline{\Gamma^D}$.

[Discrete lifting for Lagrange elements] Let U_h be a Lagrange finite element space of degree $k \geq 1$ on a mesh $\mathcal{T}_h$. For any continuous function g defined on $\overline{\Gamma^D}$, there exists a unique function $g_h \in U_h$ such that:

- $g_h(\boldsymbol{\nu}) = g(\boldsymbol{\nu})$ for every vertex $\boldsymbol{\nu}$ on $\overline{\Gamma^D}$,
- $g_h(\boldsymbol{\nu}) = 0$ for every vertex $\boldsymbol{\nu}$ not on $\overline{\Gamma^D}$.

Construction for test-case (2) In test-case (2), the Dirichlet boundary Γ^D is composed of the two vertical sides of the square $\Omega = (0,1)^2$:

$$\Gamma^D = \{0\} \times (0,1) \cup \{1\} \times (0,1).$$

The Dirichlet datum is given by:

$$u^D(0,y) = 1, \quad u^D(1,y) = e^\eta, \quad \forall y \in (0,1).$$

This function is constant on each vertical side (with respect to y), hence it is continuous on Γ^D.

The finite element space U_h is the $\mathbb{Q}^1$-conforming space defined by:

$$U_h = \{v_h \in C^0(\overline{\Omega}) \mid v_{h|K} \in \mathbb{Q}^1(K)\ \forall K \in \mathcal{T}_h\}.$$

The vertices of the mesh are the points (ih, jh) for $i, j = 0, \ldots, M$.

We define the discrete lifting $\tilde{u}_h^D \in U_h$ by specifying its values at all vertices:

- For vertices on Γ^D:
  - If $x = 0$ and $y \in [0,1]$, set $\tilde{u}_h^D(0,y) = 1$.
  - If $x = 1$ and $y \in [0,1]$, set $\tilde{u}_h^D(1,y) = e^\eta$.

- For all other vertices (interior vertices or vertices on Γ^N), set $\tilde{u}_h^D(\boldsymbol{\nu}) = 0$.

Since U_h is a Lagrange $\mathbb{Q}^1$ space, the function $\tilde{u}_h^D$ is uniquely determined by these nodal values. Moreover, by construction, the trace of $\tilde{u}_h^D$ on Γ^D coincides with u^D at the vertices. Because both u^D and $\tilde{u}_h^D$ are linear on each edge of Γ^D (as restrictions of bilinear functions to a vertical edge are linear in y), and they match at the two endpoints of each such edge, they coincide on the entire edge. Hence, $\tilde{u}_h^D$ is indeed a lifting of u^D in U_h.

2.10 Q23 : Numerical convergence study

2.10.1 Determination of Elementary Matrices using Tensor Products

In this section, we derive the elementary matrices for the $\mathbb{Q}^1$ finite element on a square cell K of side length h. Following the methodology described in the lecture notes, we utilize the tensor product structure of the $\mathbb{Q}^1$ shape functions to construct the 2D matrices from 1D reference matrices.

2.10.2 1D Reference Matrices

Let us consider a 1D reference element on the interval $[0, h]$. The linear basis functions are $\psi_0(x) = 1 - \frac{x}{h}$ and $\psi_1(x) = \frac{x}{h}$. We define the following 1D elementary matrices:

1D Mass Matrix (M_{1D}):

$$M_{1D} = \int_0^h \psi_i(x)\psi_j(x)\,dx = \frac{h}{6}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

1D Stiffness Matrix (K_{1D}):

$$K_{1D} = \int_0^h \psi_i'(x)\psi_j'(x)\,dx = \frac{1}{h}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

1D Convection Matrix (C_{1D}):

$$C_{1D} = \int_0^h \psi_j'(x)\psi_i(x)\,dx = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}.$$

2.10.3 2D Elementary Matrices construction

The 2D shape functions on the square K are defined as the tensor product of the 1D functions:

$$\varphi_I(x, y) = \psi_i(x)\psi_j(y), \quad \text{where } I \text{ maps to local indices } (i, j).$$

Consequently, the 2D elementary matrices can be assembled using the Kronecker product ($\otimes$) of the 1D matrices.

Elementary Stiffness Matrix (K_{el}) The diffusion term corresponds to the bilinear form $\int_K \nabla u \cdot \nabla v$. Using the tensor product property, the stiffness matrix decomposes into:

$$K_{el} = (K_{1D} \otimes M_{1D}) + (M_{1D} \otimes K_{1D}) .$$

Substituting the 1D expressions:

$$K_{el} = \left[\frac{1}{h}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \otimes \frac{h}{6}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\right] + \left[\frac{h}{6}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \otimes \frac{1}{h}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\right] .$$

The factors h and $1/h$ cancel out, yielding the constant matrix (independent of h):

$$K_{el} = \frac{1}{6}\begin{pmatrix} 4 & -1 & -1 & -2 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ -2 & -1 & -1 & 4 \end{pmatrix} .$$

Note: The row/column ordering here corresponds to the tensor ordering $(0,0), (0,1), (1,0), (1,1)$.

Elementary Advection Matrix (C_{el}) For the advection term with a constant velocity field $\mathbf{V} = (\eta, 0)^T$, the bilinear form is $\int_K \eta(\partial_x u)v$. This corresponds to the tensor product of the 1D convection matrix in x and the 1D mass matrix in y:

$$C_{el} = \eta\,(C_{1D} \otimes M_{1D}) .$$

Substituting the expressions:

$$C_{el} = \eta\left[\frac{1}{2}\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \otimes \frac{h}{6}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\right] = \frac{\eta h}{12}\begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \otimes \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} .$$

Calculating the Kronecker product yields:

$$C_{el} = \frac{\eta h}{12}\begin{pmatrix} -2 & -1 & 2 & 1 \\ -1 & -2 & 1 & 2 \\ -2 & -1 & 2 & 1 \\ -1 & -2 & 1 & 2 \end{pmatrix} .$$

Global System Assembly The global system matrix A is obtained by assembling the local contributions:

$$A = \sum_{K \in \mathcal{T}_h} \left(K_{el}^K + C_{el}^K\right) .$$

This algebraic formulation allows for an efficient implementation of the Finite Element Method on the Cartesian grid.

2.10.4 Numerical Results and Convergence Analysis

In this section, we analyze the convergence behavior of the Finite Element method by computing the L^2 error and the H^1 semi-norm error for different mesh sizes h and different values of the parameter $\eta \in \{1, 10, 100\}$.

The expected theoretical convergence rates for P_1 elements are $O(h^2)$ for the L^2-norm and $O(h)$ for the H^1-norm.

Case 1: $\eta = 1$

Table 4 presents the results for $\eta = 1$. The method exhibits optimal convergence behavior. The computed order for the L^2 error converges rapidly to 2.00, and the H^1 error order converges to 1.00. The absolute errors are small, indicating a stable approximation.

k	h	$\|e\|_{L^2}$	Order (L^2)	$\|e\|_{H^1}$	Order (H^1)
2	0.2500	8.63×10^{-3}	-	1.28×10^{-1}	-
3	0.1250	2.15×10^{-3}	2.00	6.44×10^{-2}	0.99
4	0.0625	5.39×10^{-4}	2.00	3.22×10^{-2}	0.99
5	0.0312	1.35×10^{-4}	2.00	1.61×10^{-2}	0.99
6	0.0156	3.40×10^{-5}	2.00	8.06×10^{-3}	0.99
7	0.0078	8.00×10^{-6}	2.00	4.03×10^{-3}	0.99

Table 4: Convergence history for $\eta = 1$.

Case 2: $\eta = 10$

Table 5 shows the results for $\eta = 10$. We observe the theoretical asymptotic convergence orders (≈ 2 for L^2 and ≈ 1 for H^1) are preserved. However, the magnitude of the errors has increased significantly compared to the $\eta = 1$ case. For $h = 0.0078$, the L^2 error is approximately 1.77, whereas it was 8×10^{-6} for $\eta = 1$. This illustrates the influence of the parameter η on the constant $C(\eta)$ in the error estimate $\|u - u_h\| \leq C(\eta)h^p$.

k	h	$\|e\|_{L^2}$	Order (L^2)	$\|e\|_{H^1}$	Order (H^1)
2	0.2500	1733.86	-	27810.42	-
3	0.1250	452.15	1.94	16546.69	0.75
4	0.0625	113.37	1.99	8721.68	0.92
5	0.0312	28.33	2.00	4422.16	0.98
6	0.0156	7.08	2.00	2218.93	0.99
7	0.0078	1.77	2.00	1110.45	0.99

Table 5: Convergence history for $\eta = 10$.

Case 3: $\eta = 100$

Table 6 presents the results for $\eta = 100$. The errors here are astronomically large (scaling up to 10^{43}). This suggests that the exact solution contains terms of the form $e^{\eta x}$, which grow exponentially.

Despite the magnitude, the L^2 convergence order tends towards 2.00 as the mesh is refined (reaching 1.98 at $k = 7$). However, the H^1 convergence is poor (ranging from 0.3 to 0.88), indicating that the mesh is not sufficiently fine to resolve the sharp gradients (boundary layers) induced by the large η. We are in a pre-asymptotic regime for the derivative approximation.

k	h	$\|e\|_{L^2}$	Order (L^2)	$\|e\|_{H^1}$	Order (H^1)
2	0.2500	3.86×10^{43}	-	3.39×10^{44}	-
3	0.1250	7.67×10^{38}	2.33	1.86×10^{44}	0.86
4	0.0625	2.97×10^{33}	1.36	1.49×10^{44}	0.32
5	0.0312	1.00×10^{28}	1.57	1.20×10^{44}	0.30
6	0.0156	2.71×10^{23}	1.88	7.69×10^{43}	0.65
7	0.0078	6.83×10^{18}	1.98	4.16×10^{43}	0.88

Table 6: Convergence history for $\eta = 100$ (Values displayed in scientific notation).

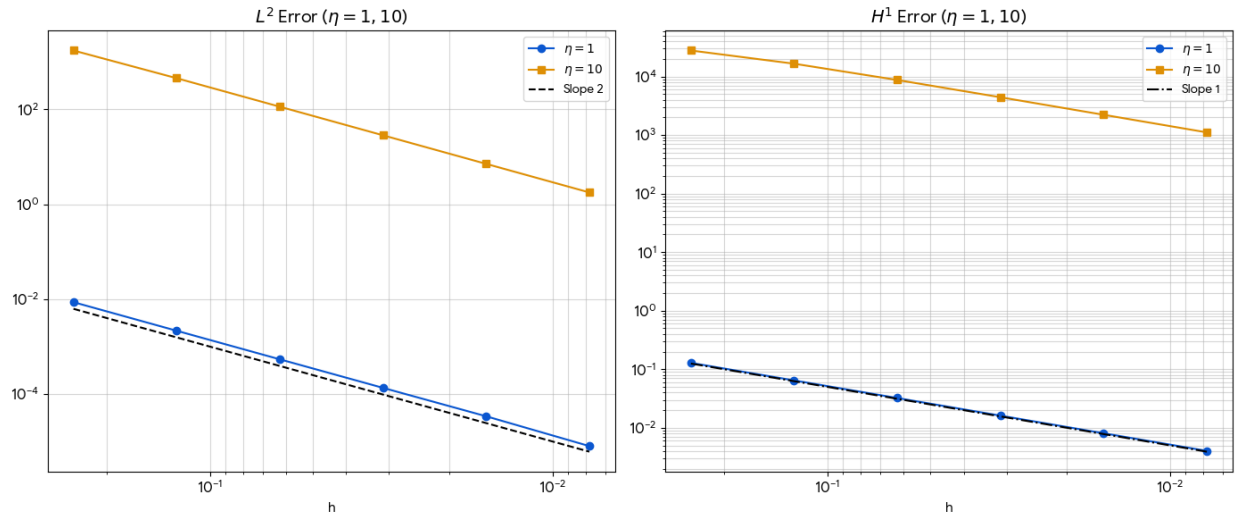

Figure 7: log-log scale errors for $\eta = 1$ and $\eta = 10$

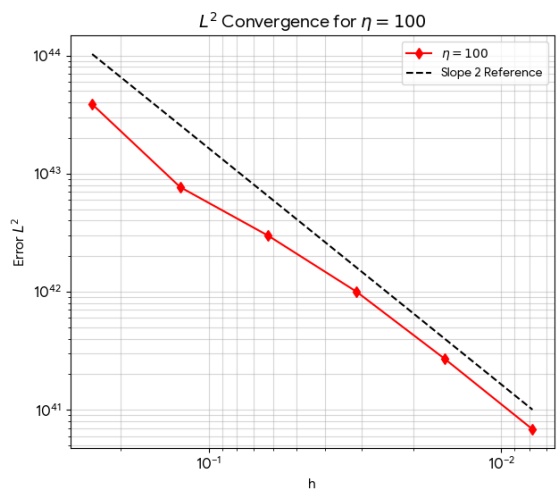

Figure 8: lo-log scale errors for $\eta = 100$

Conclusion

The numerical experiments confirm the theoretical convergence rates for low values of η. However, as η increases, the constant in the error bound grows significantly. For $\eta = 100$,

the problem becomes singularly perturbed, and extremely fine meshes are required to recover the asymptotic convergence regime, particularly for the H^1 norm.

3 BONUS

3.1 Q24 : Stability of the backward Euler scheme

The stability of Problem (12) is **not dependent** on a CFL condition.

The scheme uses the Backward Euler time discretization, which is an *implicit* method. For advection-diffusion equations, this scheme is unconditionally stable (A-stable). This means the numerical solution remains bounded regardless of the mesh size h or the time step Δt. Consequently, Δt can be chosen based solely on accuracy requirements.

3.2 Q25 : Discrete exponential convergence

Question 25: Long-Time Behavior of the Discrete Solution

We aim to prove that the discrete unsteady solution u_h^n converges to the discrete steady-state solution u_h^∞ as $n \to \infty$.

Let us define the error at time step n as:

$$e_h^n := u_h^n - u_h^\infty$$

Since both u_h^n and u_h^∞ satisfy the same Dirichlet boundary conditions on Γ^D, their difference belongs to the subspace with homogeneous Dirichlet conditions:

$$e_h^n = (\tilde{u}_h^n + \tilde{u}^D) - (\tilde{u}_h^\infty + \tilde{u}^D) = \tilde{u}_h^n - \tilde{u}_h^\infty \in U_{h,0}^D$$

1. Establishing the Error Equation The fully discrete scheme (12) at step n is:

$$\frac{1}{\Delta t}(u_h^n - u_h^{n-1}, v_h) + a(u_h^n, v_h) = (f, v_h) - a(\tilde{u}^D, v_h) \quad \forall v_h \in U_{h,0}^D \tag{3}$$

The discrete steady problem (9) satisfies:

$$a(u_h^\infty, v_h) = (f, v_h) - a(\tilde{u}^D, v_h) \quad \forall v_h \in U_{h,0}^D \tag{4}$$

Subtracting (4) from (3), and noting that $u_h^n - u_h^{n-1} = e_h^n - e_h^{n-1}$ (since u_h^∞ is constant), we obtain the error equation:

$$\frac{1}{\Delta t}(e_h^n - e_h^{n-1}, v_h) + a(e_h^n, v_h) = 0 \quad \forall v_h \in U_{h,0}^D \tag{5}$$

2. Energy Estimate We choose the test function $v_h = e_h^n \in U_{h,0}^D$. The equation becomes:

$$\frac{1}{\Delta t}\int_\Omega (e_h^n - e_h^{n-1})e_h^n + a(e_h^n, e_h^n) = 0$$

Step A: The time derivative term. Using the algebraic identity $(a-b)a \geq \frac{1}{2}(a^2-b^2)$, we have:

$$\int_\Omega (e_h^n - e_h^{n-1})e_h^n \geq \frac{1}{2}\left(\|e_h^n\|_{L^2(\Omega)}^2 - \|e_h^{n-1}\|_{L^2(\Omega)}^2\right)$$

Step B: The bilinear form term. The bilinear form is given by $a(u,v) = \int_\Omega \nabla u \cdot \nabla v + \int_\Omega (V \cdot \nabla u)v$. For any $v \in U_{h,0}^D$, the advection term satisfies:

$$\int_\Omega (V \cdot \nabla v)v = \int_\Omega V \cdot \nabla\left(\frac{v^2}{2}\right) = -\int_\Omega (\text{div } V)\frac{v^2}{2} + \int_{\partial\Omega}(V \cdot n)\frac{v^2}{2} = 0$$

This vanishes because $\text{div}(V) = 0$ in Ω, $V \cdot n = 0$ on Γ^N, and $v = 0$ on Γ^D. Thus:

$$a(e_h^n, e_h^n) = \|\nabla e_h^n\|_{L^2(\Omega)}^2$$

Using the Poincaré inequality ($\|v\|_{L^2} \leq C_P\|\nabla v\|_{L^2}$), we obtain coercivity:

$$a(e_h^n, e_h^n) \geq \frac{1}{C_P^2}\|e_h^n\|_{L^2(\Omega)}^2$$

3. Conclusion on Convergence Substituting these bounds into the error equation:

$$\frac{1}{2\Delta t}\left(\|e_h^n\|_{L^2}^2 - \|e_h^{n-1}\|_{L^2}^2\right) + \frac{1}{C_P^2}\|e_h^n\|_{L^2}^2 \leq 0$$

Rearranging the terms:

$$\|e_h^n\|_{L^2}^2\left(1 + \frac{2\Delta t}{C_P^2}\right) \leq \|e_h^{n-1}\|_{L^2}^2$$

By recurrence, we find:

$$\|u_h^n - u_h^\infty\|_{L^2(\Omega)} \leq \left(1 + \frac{2\Delta t}{C_P^2}\right)^{-n/2}\|u_h^0 - u_h^\infty\|_{L^2(\Omega)}$$

This proves that the discrete solution converges **exponentially fast** towards the discrete steady state.

3.3 Q26 : Numerical long-time behavior

In this section, we perform the numerical simulation of the unsteady problem using the Backward Euler scheme coupled with the Q^1 Finite Element Method.

Simulation Setup

We consider the test case defined by Equation (2) with the parameter $\eta = 1$. The simulation parameters are chosen as follows:

- **Mesh level:** $k = 5$, which corresponds to $M = 32$ subdivisions and a mesh size $h = 0.03125$.
- **Number of Degrees of Freedom (DoFs):** 1089.
- **Time step:** $\Delta t = 0.01$.
- **Final time:** $T = 5.0$.

At each time step n, we compute two error norms:

1. The deviation from the *discrete* steady state: $E^n_{stab} = \|u^n_h - u^\infty_h\|_{L^2(\Omega)}$.
2. The total error with respect to the *interpolated exact* solution: $E^n_{tot} = \|u^n_h - \mathcal{I}_h(u^\infty)\|_{L^2(\Omega)}$.

Results and Graphical Representation

The evolution of these two quantities is plotted in semi-logarithmic scale in Figure 9. The raw data for selected time steps is presented in Table 7.

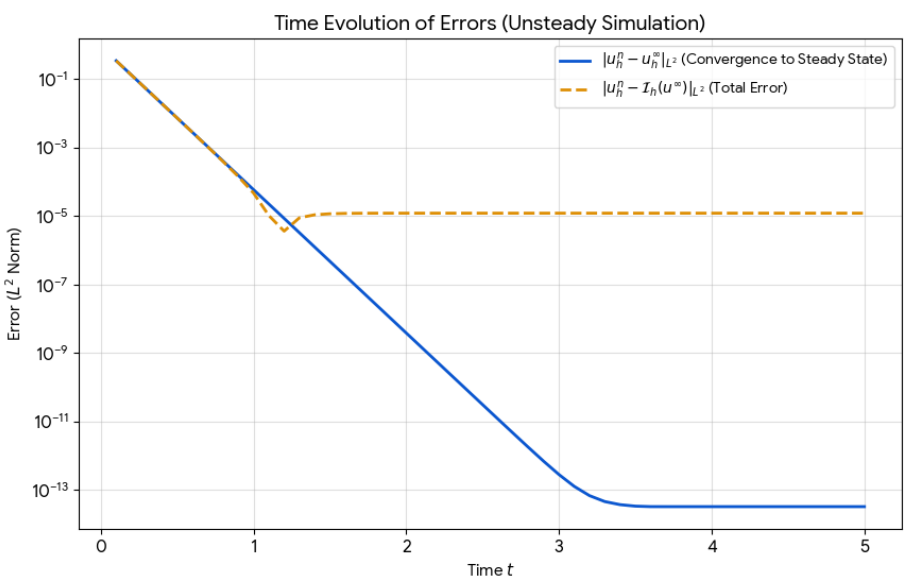

Figure 9: Time evolution of the L^2 error norms ($\eta = 1$, $h = 1/32$). Blue curve: Convergence towards the discrete steady state. Orange dashed curve: Convergence towards the exact solution (saturation).

Iteration	Time (t)	$\|u_h^n - u_h^\infty\|_{L^2}$	$\|u_h^n - \mathcal{I}_h(u^\infty)\|_{L^2}$
10	0.1	3.49×10^{-1}	3.49×10^{-1}
50	0.5	7.36×10^{-3}	7.34×10^{-3}
100	1.0	5.92×10^{-5}	4.70×10^{-5}
150	1.5	4.76×10^{-7}	1.17×10^{-5}
200	2.0	3.83×10^{-9}	1.22×10^{-5}
300	3.0	2.79×10^{-13}	1.22×10^{-5}
500	5.0	3.22×10^{-14}	1.22×10^{-5}

Table 7: Sample numerical values from the unsteady simulation.

Analysis and Comments

The results show two distinct regimes, confirming our theoretical expectations:

1. Exponential Convergence to Equilibrium (Stability) The quantity $\|u_h^n - u_h^\infty\|_{L^2}$ (solid curve) decreases linearly on the semi-log plot, which indicates an exponential decay of the form $Ce^{-\lambda t}$.

- The error decreases continuously until it hits machine precision (approx. 3×10^{-14}) around $t = 3.0$.
- This confirms the result of **Question 25**: the Backward Euler scheme is unconditionally stable and the discrete solution converges exponentially fast to the discrete steady state u_h^∞, regardless of the mesh size.

2. Saturation due to Spatial Error (Accuracy) The quantity $\|u_h^n - \mathcal{I}_h(u^\infty)\|_{L^2}$ (dashed curve) initially follows the transient decay. However, starting from $t \approx 1.5$, it saturates at a constant value of approximately 1.22×10^{-5}.

This plateau corresponds to the spatial discretization error:

$$\lim_{n\to\infty} \|u_h^n - \mathcal{I}_h(u^\infty)\|_{L^2} = \|u_h^\infty - \mathcal{I}_h(u^\infty)\|_{L^2}$$

Since u_h^∞ is only an approximation of the exact physical solution u^∞, the error cannot vanish. This residual value is controlled by the mesh size h. For Q^1 elements, this error is of order $O(h^2)$.

Conclusion The numerical scheme correctly captures the dynamics of the problem. To reduce the final error observed in the dashed curve, one would need to refine the mesh (decrease h), whereas increasing the simulation time T would have no effect.

4 Part III : General Conclusion

This project allowed us to compare different numerical strategies for solving the advection-diffusion equation. Our numerical results lead to the following conclusions:

- **Finite Volume Approximation:** The comparison between the Upwind and Scharfetter-Gummel (SG) schemes highlighted a significant qualitative difference. Our results confirmed that the **Upwind scheme** is limited to first-order convergence (≈ 0.92) and introduces substantial numerical diffusion. In contrast, the **SG scheme** demonstrated second-order convergence (≈ 2.02) and proved to be "exact" for the steady-state regime, reaching machine precision errors (10^{-11}) even on coarse meshes.

- **Finite Element Approximation (Q^1):** For low advection intensities ($\eta = 1$ and $\eta = 10$), the method behaved optimally with an $O(h^2)$ convergence rate in the L^2 norm. However, for the convection-dominated case ($\eta = 100$), our simulations revealed a strong **pollution effect**. The errors became astronomically large on coarse meshes, indicating that the standard Galerkin method requires a mesh fine enough to resolve the boundary layer (high Péclet number condition) to enter the asymptotic convergence regime.

- **Time Integration (Backward Euler):** The unsteady simulations validated the theoretical stability properties. We observed a strict **exponential convergence** towards the steady state, independent of the time step. The final accuracy was shown to be limited solely by the spatial discretization error (saturation plateau at $\approx 10^{-5}$), confirming the consistency between the dynamic and static solvers.

In summary, while standard methods (Upwind, Q^1 FEM) are robust, they lack precision in advection-dominated regimes unless the mesh is extremely fine. Structure-preserving schemes like Scharfetter-Gummel offer a much more efficient alternative for this class of problems.